

Progressive Evaluation and Enhancement of RealKeyMorph for Medical Image Registration

Alif Ilham Madani

Cornell Tech

2 West Loop Road, New York

aim57@cornell.edu

Abstract—Deformable medical image registration remains a critical challenge in clinical analysis, requiring both high spatial accuracy and parameter efficiency. This study presents a progressive evaluation and structural enhancement of the RealKeyMorph (RKM) framework. Initially, an existing RKM model, previously trained on abdominal magnetic resonance imaging (MRI) and Alzheimer’s Disease Neuroimaging Initiative (ADNI) datasets, was benchmarked against VoxelMorph, KeyMorph, and newly trained TransMorph architectures to establish baseline comparative metrics. Subsequently, we investigated RKM’s adaptability on a novel abdominal computed tomography (CT) to CT (CT-CT) dataset through extensive hyperparameter optimization, benchmarking its performance against TransMorphDiff. Observations from these evaluation phases revealed limitations in RKM’s keypoint distribution, specifically a tendency for control points to cluster. To address this, we developed and integrated a spatial dispersion loss function alongside robust data augmentation strategies. These architectural improvements were experimentally validated using a 10% subset of the ADNI dataset to facilitate rapid iteration, demonstrating significant improvements in the uniformity of the deformation field and overall registration robustness. Prior to TransMorph evaluation, detailed, modality-specific preprocessing pipelines were developed for the abdominal MRI and ADNI datasets to meet the strict standardization requirements of the transformer-based architecture.

Index Terms—Image Registration, RealKeyMorph, TransMorph, Deep Learning, Medical Imaging

I. INTRODUCTION

Deformable image registration is a fundamental process in medical image analysis, enabling the spatial alignment of anatomical structures across different imaging modalities and temporal states. While traditional iterative optimization methods provide high accuracy, they are often computationally prohibitive. Deep learning approaches, particularly resolution-agnostic models like RealKeyMorph (RKM) [1] building upon the KeyMorph framework [2], and transformer-based architectures like TransMorph [3], have significantly accelerated this process.

This study details a multiphase investigation into the RKM architecture. The research was conducted in three distinct stages: 1) A baseline comparative analysis on abdominal MRI and ADNI datasets, evaluating a pre-trained RKM model against VoxelMorph, KeyMorph, and trained TransMorph models. 2) An adaptation and optimization phase utilizing a new abdominal CT-CT dataset, where RKM’s hyperparameters were systematically tuned and evaluated against TransMorphDiff. 3) An enhancement phase that addresses identified

limitations in RKM’s keypoint extraction mechanism by introducing targeted data augmentations and a novel spatial dispersion loss function, validated on a 10% subset of the ADNI dataset.

II. METHODOLOGY

A. Datasets

To comprehensively evaluate the adaptability and structural performance of the registration frameworks, this study utilized three distinct medical imaging datasets:

1) Abdominal MRI Dataset: Sourced from the original RKM baseline study [1], this dataset consists of orthogonal 2D MRI stacks (axial and coronal) collected from 206 patients, the majority of whom present with Autosomal Dominant Polycystic Kidney Disease (ADPKD). Acquired via 1.5T and 3T scanners, the dataset features high variability in both in-plane resolution (ranging from 0.58 to 4.40 mm) and slice thickness (varying from 0.82 to 10.0 mm). From a total of 860 image pairs, 688 pairs were used for training and 172 pairs were reserved for testing. This structural heterogeneity makes it an ideal testbed for evaluating resolution-agnostic registration capabilities.

2) ADNI Brain Dataset: The Alzheimer’s Disease Neuroimaging Initiative (ADNI) provides multi-modal 3D MRI scans (including T1-weighted, T2-weighted, and FLAIR) at approximately 1 mm isotropic resolution. The full dataset comprises 2,742 training pairs and 737 testing pairs. To facilitate rapid experimental iteration during the ablation study (Phase 3), we utilized a targeted 10% subset. Specifically, 274 training pairs were sampled randomly, while 73 testing pairs were strategically sampled from the lowest 10th percentile of initial alignment quality. This intentional skewing of the test set ensured the architectural enhancements were evaluated against challenging, high-deformation edge cases.

3) Abdomen CT-CT Dataset: To assess modal adaptability, we utilized the Abdomen CT-CT dataset from the Learn2Reg challenge [4]. This dataset provides paired 3D abdominal computed tomography (CT) scans, introducing a fundamentally different intensity distribution and anatomical representation compared to MRI. It served as the primary benchmark for hyperparameter optimization and comparison against the diffeomorphic TransMorphDiff model in Phase 2, utilizing a relatively small sample size of 600 training pairs and 20 testing pairs.

B. Pipeline Architecture

Figure 1 illustrates the overall architecture of our enhanced RKM framework, highlighting our two primary structural integrations: the pre-network data augmentation pipeline and the post-extraction spatial dispersion loss.

C. Standardized Preprocessing for TransMorph

RKM requires **zero offline preprocessing**. Its resolution-agnostic architecture dynamically ingests raw medical images of varying dimensions and spacings, handling intensity scaling internally during training.

Conversely, TransMorph requires strict volumetric standardization. To establish a baseline prior to our evaluations, we implemented sequential MONAI preprocessing pipelines for both datasets (Table I).

TABLE I: Sequential Offline Preprocessing Pipelines for TransMorph

Step	Abdominal MRI Pipeline	ADNI Brain Pipeline
1	Resample to $2.5 \times 2.5 \times 5.0$ mm (bilinear/nearest)	Original 1mm isotropic preserved
2	Resize with pad/crop to $160 \times 160 \times 160$	Direct resize to $160 \times 192 \times 224$
3	Bound intensity to 0.5 th – 99.5 th percentiles	Bound intensity to 0.5 th – 99.5 th percentiles
4	Scale to $[0, 1]$ range	Scale to $[0, 1]$ range

For abdominal MRI, raw variable-resolution volumes were resampled and cropped to a strict $160 \times 160 \times 160$ grid with $2.5 \times 2.5 \times 5.0$ mm spacing. For the ADNI brain dataset, spacing standardizations were omitted. Instead, raw volumes underwent a direct interpolation resize to fit the required $160 \times 192 \times 224$ dimensions without cropping out critical cortical features. These standardizations are visually compared in Figure 2.

D. Phase 1: Baseline Evaluation on MRI and ADNI

The initial phase leveraged an RKM model that had been previously trained on the abdominal MRI and ADNI brain datasets. To establish a rigorous comparative baseline, we evaluated this against VoxelMorph, KeyMorph, and newly trained TransMorph models trained from scratch on the same datasets.

E. Phase 2: Adaptation and Optimization on Abdominal CT-CT

In the second phase, the focus shifted to evaluating the models on a novel modality: an abdominal CT-CT dataset. Concurrently, TransMorphDiff—a diffeomorphic variant of the TransMorph architecture—was trained on the abdominal CT-CT dataset to serve as a state-of-the-art comparative benchmark, prioritizing topological preservation and non-folding deformation fields.

To evaluate RKM’s adaptability to this modality, we conducted a systematic hyperparameter search across six distinct

architectural configurations (Table II). The global architecture was held constant across all configurations: a truncated U-Net (5 levels), center of mass (CoM) layer [5] for keypoint extraction, real-world coordinate alignment, and optimization using a batch size of 1 with a learning rate of 3×10^{-6} .

Training occurred in stages. All models underwent stage 1 pretraining from scratch using an affine transform and mean squared error (MSE) loss for 15,000 epochs. During stage 2 training, we systematically varied the number of extracted keypoints (64 vs. 128), the transformation strategy (affine vs. uniform thin-plate spline (TPS)), and the target loss function (Dice, structural similarity index measure (SSIM), or a combined Dice + SSIM formulation). Final evaluation (stage 3) was conducted using a combined affine + TPS transformation.

TABLE II: RKM Hyperparameter Configurations (Stage 2)

Config	Keypoints	Transform	Loss Function	Epochs
C1	128	TPS (Uniform)	Dice	20k
C2	64	Affine	SSIM	20k
C3	64	Affine	Dice + SSIM	~9k
C4	64	TPS (Uniform)	Dice + SSIM	~9k
C5	128	Affine	Dice + SSIM	~9k
C6	128	TPS (Uniform)	Dice + SSIM	~9k

F. Phase 3: Architectural Enhancements & Ablation Study

Observations across the first two phases indicated that RKM’s keypoint extractor frequently clustered points in regions of high image gradient, leaving homogenous anatomical regions with insufficient control points for accurate deformation.

To enforce uniform keypoint distribution and improve structural robustness, we introduced targeted data augmentations and a novel spatial dispersion loss. To enable rapid experimental iteration and validate these modifications efficiently, we conducted an ablation study on the 10% subset of the ADNI dataset.

The baseline model (**Experiment 1**) utilized the optimal settings derived from the original RKM architecture: 64 keypoints, TPS transformation, Dice loss, and basic intensity scaling to $[0, 1]$. Building upon this baseline, we evaluated four additional experimental configurations:

- **Experiment 2 (Spatial Augmentation):** Applied on top of the baseline, introducing random affine transformations (no scaling, $\pm 15^\circ$ rotation, ± 15 mm translation) with an 80% probability.
- **Experiment 3 (Intensity Augmentation):** Applied on top of Experiment 2, adding random gamma adjustments ($\log \gamma \in [-0.3, 0.3]$, 70% probability) and random bias fields (coefficient 0.5, order 3, 50% probability).
- **Experiment 4 (Dispersion Loss, $\lambda = 0.5$):** Applied directly to the baseline. We introduced a spatial dispersion penalty calculated on a normalized voxel grid to preserve scale and facilitate gradient propagation. Using a margin

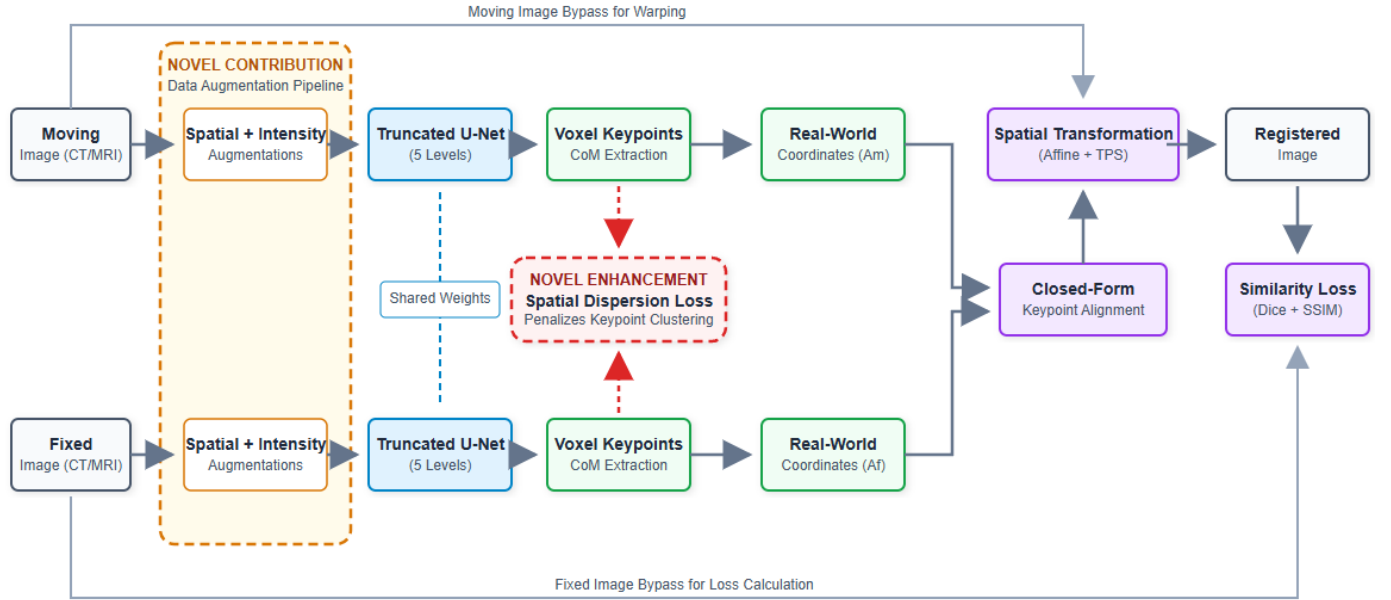


Fig. 1: Overview of the enhanced RKM architecture, highlighting the data augmentation pipeline (left) and the spatial dispersion loss module (center).

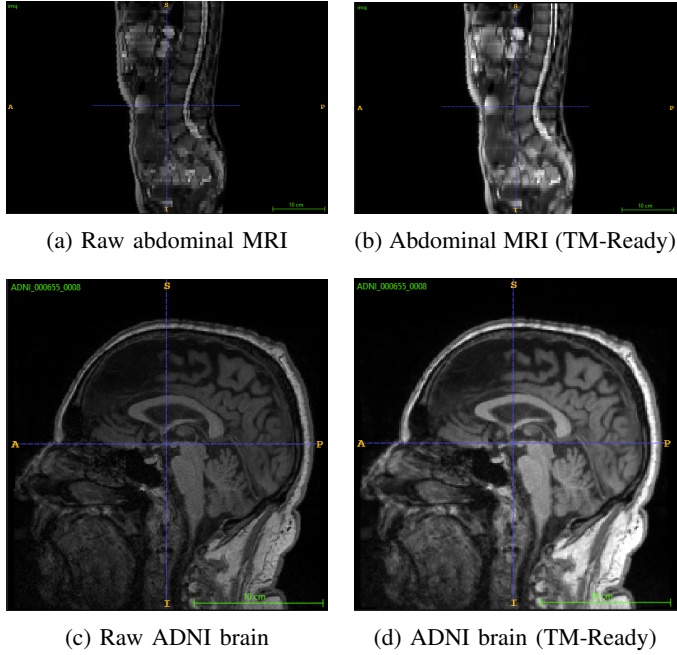


Fig. 2: Raw versus standardized inputs for the TransMorph preprocessing pipeline on abdominal MRI (top) and ADNI brain (bottom) datasets.

$m = 0.1$ defined in the normalized coordinate space $[-1, 1]^3$, the loss penalizes keypoints closer than m :

$$L_{\text{disp}} = \frac{1}{N(N-1)} \sum_{i \neq j} \max(0, m - \|k_i - k_j\|_2) \quad (1)$$

The total loss averages the moving and fixed dispersion

penalties:

$$L_{\text{total}} = L_{\text{dice}} + \lambda_{\text{disp}} \left(\frac{L_{\text{disp_moving}} + L_{\text{disp_fixed}}}{2} \right) \quad (2)$$

with the dispersion weight set to $\lambda_{\text{disp}} = 0.5$.

- **Experiment 5 (Dispersion Loss, $\lambda = 50$):** Identical to Experiment 4, but the dispersion weight was drastically increased to $\lambda_{\text{disp}} = 50$ to aggressively enforce keypoint separation.

To quantitatively assess the efficacy of the spatial dispersion loss, the spatial distribution of the extracted keypoints was evaluated using three geometric metrics: mean nearest neighbor distance (MNND) to measure the degree of local point clustering, convex hull volume to evaluate the global anatomical coverage of the keypoints, and total spatial variance to assess the overall spread of the points from their geometric center.

III. RESULTS AND DISCUSSION

A. Baseline Results on MRI and ADNI

The comparative analysis between the pre-trained RKM and baseline models on the abdominal MRI and ADNI brain datasets is detailed in Table III. RKM demonstrates exceptional efficiency, relying on only 64 keypoints while significantly outperforming VoxelMorph and the dense 512-keypoint KeyMorph baseline.

As shown in Table III for the abdominal dataset, RKM (affine) achieves the lowest Hausdorff distance (HD) (23 mm) and matches the top median Dice score (79%) with the tightest variance (7% IQR). RKM (TPS) achieves the highest SSIM (64%). For the brain dataset, RKM (affine) again leads in structural overlap, achieving the highest Dice

TABLE III: Similarity Metrics for RKM and Baseline Models, Median (IQR).

Dataset & Model	Keypoints	Dice ($\uparrow$)	HD (mm) ($\downarrow$)	SSIM ($\uparrow$)
<i>Abdominal MRI</i>				
VoxelMorph Baseline	Dense	0.54 (0.44)	49.0 (12.0)	0.55 (0.10)
KeyMorph Baseline	512	0.51 (0.40)	99.0 (40.0)	0.44 (0.14)
TransMorph	Dense	0.70 (0.25)	45.0 (32.0)	0.58 (0.16)
TransMorphDiff	Dense	0.79 (0.17)	33.0 (28.0)	0.54 (0.17)
RealKeyMorph (Affine)	64	0.79 (0.07)	23.0 (10.0)	0.60 (0.10)
RealKeyMorph (TPS)	64	0.75 (0.15)	24.0 (12.0)	0.64 (0.12)
<i>Brain (ADNI)</i>				
VoxelMorph Baseline	Dense	0.63 (0.19)	67.0 (19.0)	0.55 (0.20)
KeyMorph Baseline	512	0.57 (0.17)	70.0 (20.0)	0.43 (0.20)
TransMorph	Dense	0.81 (0.18)	3.8 (4.4)	0.72 (0.15)
TransMorphDiff	Dense	0.81 (0.07)	3.66 (1.32)	0.71 (0.16)
RealKeyMorph (Affine)	64	0.90 (0.04)	49.0 (13.0)	0.79 (0.13)

(90%) and SSIM (79%). However, TransMorphDiff captures global boundaries more accurately in the brain space, yielding the lowest HD (3.66 mm). Overall, RKM delivers highly competitive structural accuracy compared to dense, heavily preprocessed transformer architectures.

B. Abdominal CT-CT Performance

On the abdominal CT-CT dataset, the hyperparameter-optimized RKM configurations demonstrated distinct performance trade-offs based on the applied loss functions and transformations. As detailed in Table IV, training solely with Dice loss (**Configuration 1**) yielded the highest Dice score (0.338 ± 0.042), though it resulted in higher MSE and lower SSIM compared to mixed-loss formulations. Conversely, optimizing purely for SSIM (**Configuration 2**) maximized structural similarity (0.534 ± 0.036) but severely degraded the Dice score (0.191 ± 0.060).

TABLE IV: RKM Hyperparameter Optimization Results on Abdominal CT-CT

Config.	Dice ($\uparrow$)	SSIM ($\uparrow$)	MSE ($\downarrow$)	HD (mm) ($\downarrow$)
C1	0.338 $\pm$ 0.042	0.471 $\pm$ 0.039	0.028 $\pm$ 0.007	93.96 $\pm$ 7.83
C2	0.191 $\pm$ 0.060	0.534 $\pm$ 0.036	0.022 $\pm$ 0.007	88.31 $\pm$ 15.29
C3	0.313 $\pm$ 0.018	0.485 $\pm$ 0.041	0.026 $\pm$ 0.007	91.13 $\pm$ 11.77
C4	0.308 $\pm$ 0.027	0.492 $\pm$ 0.034	0.025 $\pm$ 0.006	90.54 $\pm$ 11.41
C5	0.303 $\pm$ 0.017	0.492 $\pm$ 0.035	0.025 $\pm$ 0.006	84.84 $\pm$ 15.80
C6	0.306 $\pm$ 0.028	0.493 $\pm$ 0.034	0.025 $\pm$ 0.006	92.83 $\pm$ 8.05

TABLE V: Benchmark Comparison on Abdominal CT-CT Dataset (Mean $\pm$ Std)

Model	Dice Score ($\uparrow$)	HD (mm) ($\downarrow$)	SSIM ($\uparrow$)
RealKeyMorph (Config 1)	0.338 $\pm$ 0.042	93.96 $\pm$ 7.83	0.471 $\pm$ 0.039
TransMorphDiff	0.400 $\pm$ 0.050	38.38 $\pm$ 7.11	0.650 $\pm$ 0.050

Because Configuration 1 achieved the highest structural overlap (Dice) among the RKM variants, it was selected as the representative baseline to compare against the diffeomorphic state-of-the-art model, TransMorphDiff (Table V). Despite being RKM’s best-performing configuration for overlap, it exhibited a noticeable performance gap on this novel modality. TransMorphDiff achieved superior spatial accuracy with a

mean Dice score of 0.400 ± 0.050 , a significantly lower HD of 38.38 ± 7.11 mm, and a higher SSIM of 0.650 ± 0.050 .

This quantitative delta highlights RKM’s struggle with keypoint distribution on the CT modality prior to the implementation of the spatial dispersion loss and augmentation pipeline. Furthermore, this underperformance can be heavily attributed to the limited size of the Abdomen CT-CT dataset (only 600 training pairs). Resolution-agnostic keypoint extraction inherently requires massive data diversity to learn robust, modality-invariant geometric anchors; the lack of sufficient training examples hindered RKM’s ability to generalize across the highly deformable CT structures compared to the densely supervised TransMorphDiff model.

C. Impact of RKM Enhancements (Ablation Study)

The quantitative results of the architectural ablation study—evaluated on the 10% subset of the ADNI dataset—are detailed in Figure 3.

The most significant performance gains in structural alignment were observed in **Experiment 3 (Intensity Augmentation)**, which achieved the highest overall structural overlap (**Figure 4**). By introducing random gamma and bias field augmentations, the model’s Dice score peaked at 0.820 ± 0.034 under TPS transformation ($\lambda \geq 0.1$), demonstrating that intensity invariance is a critical factor for robust keypoint extraction in this modality. Conversely, pure spatial augmentations (**Experiment 2**) yielded results identical to the baseline, indicating the model was already spatially robust.

The introduction of the spatial dispersion loss (**Experiments 4 and 5**) successfully penalized keypoint clustering while maintaining competitive structural accuracy (Dice ~ 0.740 with TPS). To visually and quantitatively validate this improvement in keypoint distribution, we computed spatial statistics for the extracted control points (Table VI) and visualized their 3D coordinates (Figure 5).

TABLE VI: Quantitative Evaluation of Keypoint Spatial Dispersion

Configuration	MNND ($\uparrow$)	Hull Volume ($\uparrow$)	Total Variance ($\uparrow$)
Baseline RKM	0.0882	0.3098	0.2007
Enhanced ($\lambda_{\text{disp}} = 0.5$)	0.1190	0.3677	0.1979
Aggressive ($\lambda_{\text{disp}} = 50$)	0.1463	0.4507	0.2139

The baseline model exhibited an MNND of 0.0882, reflecting severe localized clustering. The introduction of the spatial dispersion loss ($\lambda_{\text{disp}} = 0.5$) increased the MNND to 0.1190 and expanded the convex hull volume by 18.7%, demonstrating a more uniform distribution of control points across the anatomical space. Under aggressive penalization ($\lambda_{\text{disp}} = 50$), the hull volume expanded by 45.5% relative to the baseline. Notably, total spatial variance remained relatively stable across all configurations, indicating that the dispersion loss successfully forces keypoints apart without causing them to unproductively diverge beyond the boundaries of the anatomical structure.

However, aggressively scaling the dispersion penalty to $\lambda = 50$ (**Experiment 5**) introduced instability during the rigid

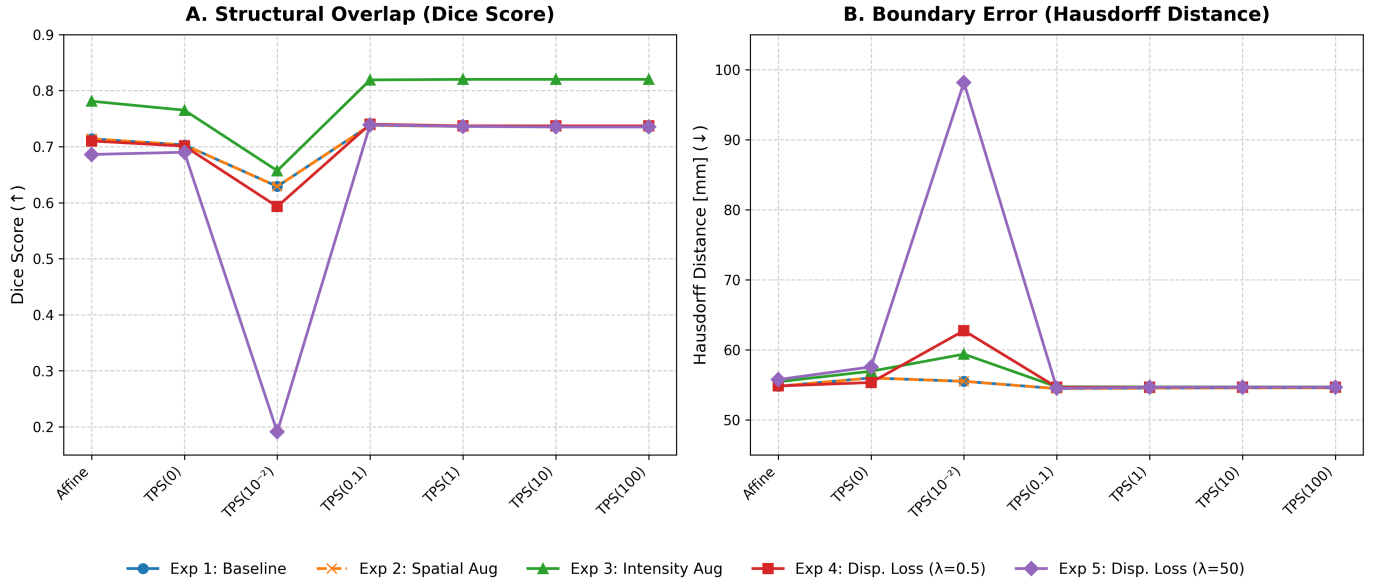


Fig. 3: Ablation study results evaluating structural overlap (A) and boundary error (B) across varying transformation stages and TPS smoothing weights (λ).

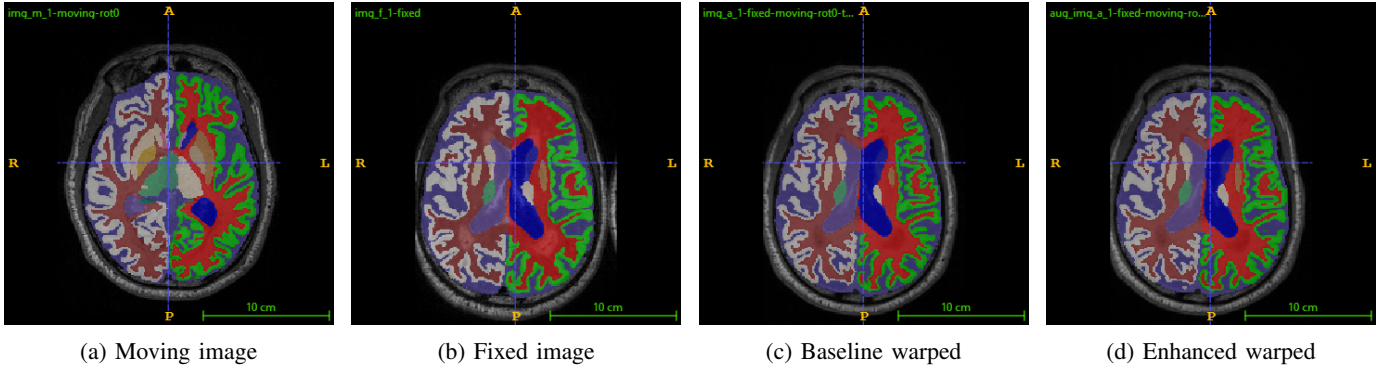


Fig. 4: Qualitative comparison of multimodal registration on the ADNI brain dataset for baseline and augmented models.

affine stage, dropping the baseline Dice to 0.686 ± 0.075 , though the non-linear TPS stage was able to recover the alignment.

Across all experimental configurations, an explicit trend emerged regarding TPS regularization: a low smoothing weight ($\lambda = 10^{-2}$) consistently led to severe deformation folding and metric collapse (e.g., HD spiking to 98.16 ± 18.57 mm in Experiment 5), whereas values of $\lambda \geq 0.1$ effectively stabilized the deformation field.

IV. CONCLUSION

This study provides a comprehensive evaluation and progressive refinement of the RKM architecture. By establishing baseline comparisons with TransMorph and other established frameworks on MRI and ADNI datasets, and subsequently optimizing RKM against TransMorphDiff on a novel abdominal CT-CT dataset, we identified specific areas for architectural improvement. The introduction of data augmentations and a spatial dispersion loss effectively mitigated keypoint clustering

issues, significantly enhancing the model’s robustness and deformation field quality. These findings demonstrate that with targeted enhancements, resolution-agnostic frameworks like RKM can achieve highly competitive parameter efficiency while utilizing sparse keypoints, offering notable advantages over dense, voxel-wise transformer methodologies.

Future work will focus on three primary avenues to further refine the RKM framework. First, we aim to integrate mask guidance using segmentation labels, penalizing keypoints that map to background or empty spaces while explicitly rewarding those positioned within anatomical regions of interest. Second, we plan to combine this mask guidance directly with the proposed spatial dispersion loss to ensure keypoints are both uniformly distributed and anatomically relevant. Finally, we will investigate multi-scale keypoint extraction by integrating CoM layers at various U-Net resolutions. Deep layers can extract globally-aware anatomical anchors with large receptive fields, while shallow layers can extract precise, localized

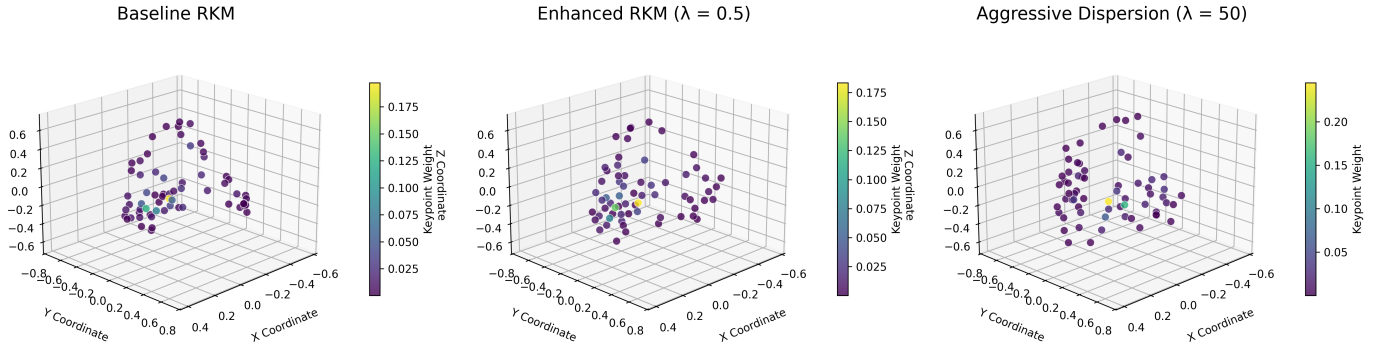


Fig. 5: 3D spatial distribution of extracted keypoints under varying spatial dispersion penalties.

boundary points. Feeding this diverse set of control points into the closed-form transformation solver could improve the topological stability of the non-linear deformation field while maintaining parameter efficiency.

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